

Impact Data and Evidence Aggregation Library (IDEAL)

Data extraction principles and objectives

The Impact Data and Evidence Aggregation Library (IDEAL) is a collaborative initiative to create a comprehensive, open-source and open-access repository of randomized controlled trials (RCTs) conducted in low- and middle-income countries across disciplines. IDEAL systematically documents impact evaluation studies to support evidence synthesis, meta-analysis, and evidence-informed policy and practice. This document describes IDEAL's metadata schema, the survey, data extraction process, and quality assurance processes, providing researchers, policymakers, reviewers, and potential partners with a comprehensive understanding of how IDEAL operates.

1. Overview of IDEAL

IDEAL functions as a neutral metadata repository rather than a standards-based review system. It systematically captures detailed information about the RCTs being evaluated including interventions, study arms, design features, and results, while leaving users free to apply their own quality standards and analytical frameworks. Rather than rating studies—as review systems like the What Works Clearinghouse do—IDEAL collects the raw information that enables secondary users to conduct their own assessments according to their preferred methodologies. This descriptive approach reflects IDEAL's cross-disciplinary scope, recognizing that the relevance of specific validity considerations varies across fields and that researchers may wish to apply different quality frameworks appropriate to their research questions.

Objectives and Principles. IDEAL is guided by four core principles that shape all aspects of the library's design and implementation:

- **Transparency:** IDEAL operates as an open-source, open-access global public good. All methodological decisions, coding protocols, and quality assurance are documented and publicly available. The library provides clear information about what data is collected, how it is extracted, and what standards are applied, enabling users to understand, use, and appropriately interpret IDEAL's data.
- **Reproducibility:** IDEAL's processes and outputs are designed to be replicated and used by third-parties. Detailed protocols document each step of the coding process, along with an open-source data entry mask. The open-source approach ensures that partner organizations can adopt and adapt IDEAL's methods fully or even partially, and that the broader research community can verify and build upon IDEAL's work.
- **Minimum set of fields:** Rather than attempting to capture every possible detail about each study, IDEAL focuses on a carefully selected minimum set of fields essential for evidence synthesis and risk-of-bias assessment. This approach reflects the library's multi-disciplinary scope, recognizing that the relevance and importance of specific information can vary across disciplines and research domains. In addition, it balances comprehensiveness with feasibility, enabling systematic coding at scale while capturing the most critical information for users to conduct meta-analyses, systematic reviews, and evidence-based decision-making.
- **Scalability and sustainability:** IDEAL's design prioritizes approaches that can be implemented efficiently across large numbers of studies. The staged workflow, quality control procedures, and technology platform are all structured to support systematic data extraction from thousands of RCTs by various users while maintaining high data quality.

IDEAL Collaboration. IDEAL is a collaborative initiative launched in June 2023, bringing together member and partner organizations committed to building open-access evidence infrastructure for

development research. The initiative operates through a technical working group consisting of Principal Investigators from IDEAL's member and partner organizations, who meet regularly to develop the library's methodological outputs. IDEAL is an ongoing initiative that welcomes new collaborators constantly. The leading organization is the Strategic Impact Evaluation Fund at the World Bank. IDEAL has benefited from individuals at the Development Research Group and the Development Impact Group at the World Bank, the Center for Effective Global Action at the University of California, Berkeley, AidGrade, Northwestern University, Innovations for Poverty Action, University of California, Los Angeles, and the International Initiative for Impact Evaluation (3ie).

This collaborative model reflects IDEAL's commitment to building sustainable, distributed capacity for evidence synthesis. By training coders and supervisors across multiple organizations and establishing shared standards and protocols, IDEAL aims to create an infrastructure that can scale efficiently while maintaining quality.

IDEAL Outputs. IDEAL's database infrastructure is being developed across four main outputs:

1. **Schema and extraction protocol:** IDEAL's metadata schema defines the conceptual framework for what information is extracted from each study. This conceptual schema is operationalized through structured data extraction instruments, which translate the schema into specific fields, response options, validation rules, and skip logic. Together, these components enable consistent and systematic data extraction at scale.
2. **Data entry platform.** IDEAL developed a custom, public, and free data entry platform to systematically collect data from papers using its survey fields. The platform integrates data entry mask, quality measurement procedures, and support systems to provide a one-stop solution for data extraction. The platform also has the functions of a general survey tool and can be used by any users to create their own surveys outside of IDEAL.
3. **Library of RCTs:** IDEAL is building a comprehensive library of randomized controlled trials conducted in low- and middle-income countries. Each study in the library is coded using the standardized schema, creating consistent, comparable information across all studies. The library is designed to be interoperable, able to pull information from other platforms that use the same schema.
4. **Access platform:** IDEAL is developing a platform that will allow researchers to download extracted data for their own analyses and presents information in accessible and interactive ways for non-technical audiences including policymakers and practitioners.
5. **Pedagogical resources:** To support broader adoption of evidence synthesis methods and transparency about IDEAL's approach, IDEAL is developing comprehensive pedagogical resources. These include training courses and technical materials for coders and external users, documentation of IDEAL's methodology and standards (including this review guide), and open-source data files that explain the assumptions underlying standard packages for meta-analysis in statistical software. These resources aim to mainstream evidence aggregation methods among researchers, practitioners, and policymakers for better knowledge and decision making.

Meta-data Schema

IDEAL aims to present consistent information on treatment effects from RCTs conducted in the social sciences in low- and middle-income countries. While the goal is not to eliminate the need for meta-analytic researchers to read original studies, IDEAL will always present effect sizes that have been standardized across studies and information that will permit library users to restrict their comparisons by study attributes, such as context, experimental design, measurement, and intervention design and to understand what information relevant for evidence aggregation may be contained in the study, such as the estimation of heterogeneous treatment effects.

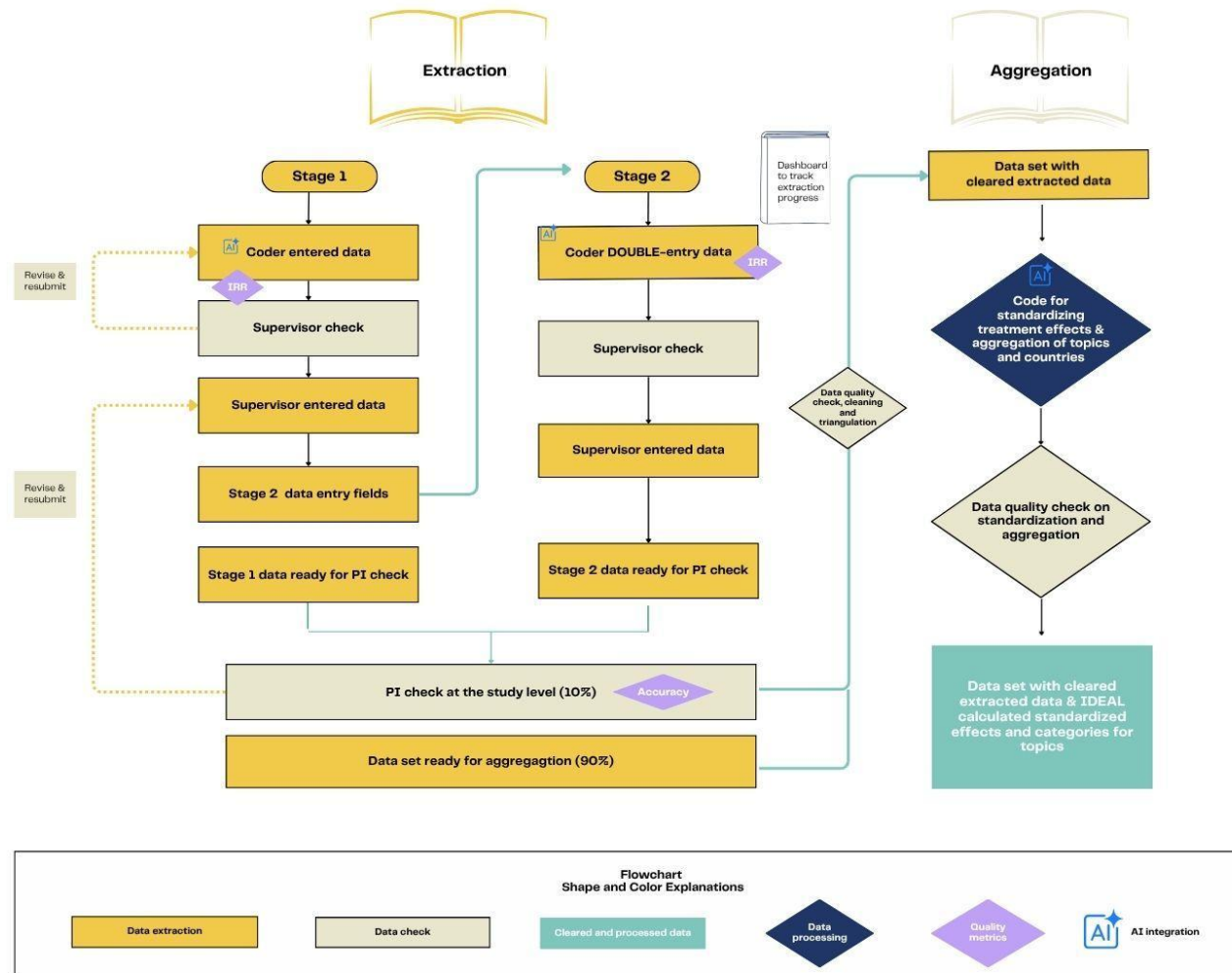
To construct a set of minimum fields, the IDEAL team first conducted a desk-review, consulting the data collection tools used by researchers who had conducted meta-analysis, the fields used by existing evidence repositories (such as AidGrade and ClinicalTrials.gov), and guidelines used in evidence aggregation, such as PRISMA and GRADE. Next the team consulted external experts including more than 20 specialists in social sciences, evidence aggregation, and meta-analysis who participated in external working group meetings in September and October 2023. The meta-data schema was ultimately approved by the IDEAL Steering Committee in November 2024.

The Survey

To operationalize the data extraction for the minimum set of fields in the meta-data schema, the IDEAL team has developed a set of survey fields to capture relevant information from each individual paper. During the pilot phase, human coders are testing survey fields on 1,000 papers. Data was first extracted using a survey mask built with Open Data Kit (ODK) tools on SurveyCTO and is now using an IDEAL developed custom data entry website. The initial dataset coded and checked by humans will serve as the ground truth data for future automated data extraction tools that would be integrated into IDEAL.

Staged Data Extraction Workflow. IDEAL employs a two-stage data extraction workflow designed to manage complexity, ensure quality, and create logical dependencies between different types of information (Figure 1):

Figure 1: IDEAL Data Extraction Workflow



- **Stage 1** extracts the structural characteristics of experiments, identifying how many experiments a paper reports, what interventions and study arms exist, which outcomes have treatment effects, what empirical specifications are used, when data was collected, and which exhibits (tables, figures, text) contain results. This structural mapping determines what questions appear in later stages.

It also systematically locates treatment effects by matching outcomes, arm comparisons, and specifications with the actual results reported in each exhibit. IDEAL's prioritization hierarchy guides this process, ensuring that the most analytically appropriate estimates are identified for each outcome while also documenting author-preferred specifications when they differ.

- **Stage 2** collects comprehensive details about experiments, interventions, outcomes, samples, and the identified treatment effects. This includes intervention descriptions, outcome definitions, sample characteristics, implementation context, and the numerical values and precision measures for treatment effects and the additional statistics needed to standardize them as effect sizes.

Quality checkpoints between stages ensure that only validated information flows forward, with supervisor review after Stages 1, and double-coding employed in Stage 2 where the bulk of detailed information is extracted.

Roles and Responsibilities. IDEAL's quality assurance system depends on clearly defined roles:

- **Coders** extract information from published papers following detailed protocols and training. Coders complete the two stages for assigned papers, working systematically through the coding survey and incorporating supervisor feedback to continuously improve their work.
- **Supervisors** review coding submissions after Stage 1, selecting or correcting answers to create validated datasets for subsequent stages, and providing detailed feedback to coders. Supervisors also reconcile discrepancies between the two independent coders in Stage 2 to produce final consensus data. Supervisors also host regular office hours to answer questions from coders.
- **Principal Investigators (PIs)** conduct spot checks on a subset of supervisor-reviewed papers (i.e. 20% initially and then 10%), provide additional quality oversight, establish ground truth for training papers, and make final decisions on methodological questions or complex coding issues that arise during implementation.
- **Data management team** monitors the data extraction process by tracking progress, maintaining the data entry mask and processing submitted data. The management team organizes regular check-ins with coders, supervisors and PIs to provide updates on technical and process issues in data extraction.

Using This Guide. This guide is organized into six main sections following this overview:

- **Section 2** describes the screening criteria that determine which studies are eligible for IDEAL review
- **Section 3** defines what constitutes an "experiment" in IDEAL's framework and how to identify single versus multiple experiments
- **Section 4** details the core components IDEAL extracts from each experiment and why this information matters
- **Section 5** explains the two-stage data extraction workflow and how information flows through the system
- **Section 6** describes IDEAL's quality assurance procedures, training requirements, and reliability metrics

Together, these sections provide a complete picture of how IDEAL transforms published experimental research into a systematic, searchable database of evidence.

2. Screening & Scope

The IDEAL Library applies systematic screening criteria to identify eligible studies for inclusion. This section describes the specific requirements studies must meet to be considered for data extraction. Those criteria ensure that the library maintains methodological consistency and includes only those studies that meet minimum eligibility and relevance thresholds.

2.1 Study Eligibility Requirements

Eligible research report. To be eligible for IDEAL data extraction in the pilot phase¹, studies must meet the following requirements.

- **Peer-Reviewed journal articles.** Studies must be published in a peer-reviewed journal to be eligible for IDEAL inclusion. Peer review provides an independent assessment of methodological quality and ensures that studies have undergone systematic evaluation by subject matter experts prior to publication. This requirement ensures a baseline level of methodological scrutiny, stability of results, and accessibility for the initial library development. Future phases may expand to include other publication venues and formats.
- **Availability.** Studies must have a DOI to guarantee permanent access. This requirement ensures that IDEAL library users can access and review the original studies and facilitates transparency in the coding process.
- **Timeframe** Studies published from [1980] onward are eligible.
- **Language.** Studies must be available in English.

Eligible countries and topics. Studies must be implemented in **low- and middle-income countries**. IDEAL defines low- and middle-income countries according to [World Bank income group classifications](#) at the time of the study's implementation and make annually adjustments. This geographic restriction focuses the library on contexts most relevant to international development policy and practice, while recognizing the distinct challenges and opportunities present in resource-constrained settings. Studies conducted entirely in high-income countries are not eligible, regardless of their methodological rigor or policy relevance.

Eligible study topics. At least one intervention or one outcome of the study is in the realm of social sciences. IDEAL adopts a **cross-disciplinary approach** to topical coverage, encompassing field experiments across diverse substantive areas including but not limited to education, health, governance, agriculture, labor markets, finance, environment, and social protection. This broad scope reflects the library's goal of providing comprehensive documentation of experimental evidence relevant to development policy and practice.

Eligible study design. The study design must be based on **randomized controlled trials (RCTs)** that evaluate field or policy interventions (e.g., field experiments, A/B tests, cluster-randomized trials). Random assignment to treatment and control conditions provides the most rigorous approach to causal inference. The sole focus on RCT also increases the comparability across studies in IDEAL.

Exclusions. Some experimental studies are excluded from the IDEAL review even if they employ random assignment.

¹ In the pilot phase, IDEAL aims to code 1,000 RCTs by human coders. The eligibility criteria may change after IDEAL completes its pilot.

- Lab-in-the-field experiments are not eligible for inclusion. While these studies may use randomization and occur in field settings, they typically involve artificial decision-making tasks or controlled experimental environments that differ substantively from policy-relevant field interventions.
- Design interventions are excluded from the library. These studies randomize the order, framing, or wording of survey questions to examine measurement effects rather than testing substantive policy or programmatic interventions. These experiments fall outside the scope of policy-relevant field trials that IDEAL aims to catalog.
- Hypothetical experiments are not eligible. Studies must report results from interventions that were actually implemented. Proposed experiments, pilot protocols without results, pre-analysis plans or simulations of potential interventions do not meet the library's implementation requirement.

Eligible treatment effects reporting. To be eligible for inclusion, studies must report **at least one quantitative treatment effect** with available data on magnitude of the effect and statistical precision for standardization. This information may appear in tables, figures, or text within the manuscript. The estimated treatment effects needs to have A numeric estimate of the contrast between two groups, the actual magnitude, for example difference in means, difference in proportions, odds or hazard ratio, etc., as well as sufficient information to recover precision, for example SE, CI, p-value, or test statistic. With more than two arms, at least one pairwise comparison is required, an omnibus test across arms does not qualify. Effects estimated as slopes on continuous variables do not qualify, including treatment interacted with a continuous characteristic such as income. Effects reported separately for defined subgroups do qualify, such as a treatment-control difference reported separately for girls and for boys, or for older and younger children.

Studies that report only descriptive statistics, qualitative findings, or treatment effect estimates without precision data do not meet this minimum reporting threshold. This requirement ensures that included studies contribute quantitative evidence on intervention impacts that can inform future meta-analyses and evidence synthesis efforts. Studies may report effects on any outcome domain relevant to the intervention.

3. Experiments in IDEAL: Definitions and Scope

IDEAL records published manuscripts that report results from randomized controlled trials. The coding unit is “experiment” (or “trial”). Within each manuscript review, IDEAL identifies and codes one or more experiments being evaluated. This section defines what IDEAL considers an "experiment," provides guidance for determining whether a manuscript contains a single experiment or multiple experiments, and describes the intervention assignment strategies such as parallel, factorial, crossover, and adaptive designs that IDEAL documents for each experiment. Additionally, because published papers vary widely in their complexity, some reporting a single straightforward trial while others presenting multiple related or independent experiments with sophisticated randomization schemes, this section also provides information on how IDEAL handles those cases. Such information is critical for users conducting meta-analyses, as they affect sample independence and appropriate statistical methods for evidence synthesis.

3.1 Defining an Experiment

For the purposes of IDEAL coding, an **experiment** is defined as a **unified research design** that uses randomization to create a valid counterfactual to evaluate the causal effects of one or more interventions on a **defined sample**.

The critical question in determining whether a paper contains one or multiple experiments is: Were the randomization procedures designed and implemented as parts of an integrated research design, or do they represent independent studies?

When Does a Manuscript Contain Multiple Experiments?

A manuscript contains multiple experiments when it reports results from **fundamentally separate research designs** that operate independently. Key indicators of multiple experiments include:

- **Independent samples:** Different populations or geographic areas were sampled and randomized separately. For example, a manuscript reporting an education RCT conducted in 100 Kenyan schools and a separate health RCT conducted in 50 Tanzanian villages contains two experiments.. [Barrera-Osorio et al. \(2011\)](#) conducted two experiments to evaluate different designs of conditional cash transfer interventions in two localities: San Cristobal and Suba.
- **Unrelated interventions tested separately:** Different interventions were evaluated through separate, non-integrated randomization schemes. For example, if a manuscript reports that 80 schools were randomized to teacher training versus control in 2015, and separately reports that 60 different schools were randomized to student tutoring versus control in 2017, these are two distinct experiments.
- **Sequential studies without integration:** The manuscript reports results from studies conducted at different times with different samples, even if testing similar interventions. While results may be presented together for comparison, the underlying experiments were independent.

Exclusions When a manuscript includes both eligible field experiments and ineligible designs (such as lab-in-the-field experiments or a hypothetical intervention), only the eligible field experiment(s) are coded.

Many experiments employ sophisticated designs involving multiple stages or dimensions of randomization. These complex designs remain single experiments when they are part of an integrated research plan. Section 3.2 describes how IDEAL classifies these designs.

3.2 Intervention Assignment Strategies and Multi-Stage Randomization

Within each experiment, IDEAL documents the overall design structure that determines how interventions are allocated to study arms. Understanding these assignment strategies is important for interpreting results and conducting appropriate statistical analyses. Some experiments also use multi-stage randomization, assigning interventions at more than one level or time point; this feature is documented separately and can occur within any assignment strategy.

IDEAL classifies intervention assignment strategies into the following categories:

- **Parallel Design:** The most common strategy in randomized controlled trials. Each intervention is assigned to only one arm, creating distinct groups that can be compared (e.g., treatment vs. control, or multiple treatment arms with different interventions)..
 - Example 1: Schools are randomized into three distinct groups, recognition program, in-kind performance rewards, and control group, as in [Barrera-Osorio et al, 2022](#).
- **Factorial Design.** Used when evaluating two or more interventions both alone and in combination. At least one intervention is assigned to more than one study arm, allowing researchers to test individual effects and interaction effects between interventions

- Example: Four groups are created, psychosocial stimulation only, micronutrient supplementation only, both interventions combined, and control (neither intervention) as in Andrew et al, 2018.
- **Crossover Design.** Each study arm receives different interventions (including no intervention) at different phases of the study. This category also includes phase-in or stepped-wedge designs where intervention rollout is randomized and every unit ultimately receives the program. **Important note:** If the study's endline data collection occurs before units receive interventions beyond their initial assignment, IDEAL does not classify the design as crossover but rather as parallel design with staggered implementation.
 - Example: Doctors and patients switch between control and treatment conditions on different days, creating within-unit comparisons, Lopez et al, 2022.
- **Other.** For assignment strategies that don't fit the above categories. When this classification is used, IDEAL coders specify the nature of the design in detail. IDEAL may not be able to record papers using assignment strategies other than parallel, factorial and cross-over.²

Separately from assignment strategy, IDEAL documents whether an experiment involves more than one distinct randomization event, occurring at different units of randomization or at different points in time.

Multi-Stage Randomization. A multi-stage design randomly assigns interventions through two or more distinct randomization events, which may occur at different units of randomization or at different points in time. For example, a study may first randomize districts to treatment or control, then randomize units within treated districts to receive the intervention. These randomizations may occur simultaneously or sequentially over the course of the study. This design feature is independent of the intervention assignment strategy (parallel, factorial, cross-over) and is recorded separately in IDEAL.

- Example: Ichino and Schündeln (2012) used a two-stage randomized design. In the first stage, constituencies were randomly assigned to treatment or control. In the second stage, electoral areas within each treated constituency were randomly selected to receive the intervention. This study involved two units of randomization (constituencies and electoral areas) and would be coded as multi-stage.

3.3 Multiple Manuscripts from One Experiment

Multiple manuscripts may present results from a single experiment. IDEAL codes these manuscripts separately but identifies them as **related studies** ex post to inform users that the samples are not independent. Common scenarios include:

- **Different follow-up periods:** One manuscript reports short-term effects at 12 months while another reports long-term effects at 36 months of the same participants receiving the same intervention. These manuscripts present different temporal analyses of one experiment.
- **Different outcomes:** One manuscript reports education outcomes while another reports health outcomes from the same experimental sample and intervention. These are different manuscripts analyzing different outcomes from one experiment.

² For example, another assignment strategy is adaptive design where the randomization rule can change during the trial based on experimental data collected during the study. Assignment probabilities may be modified based on interim results from previous waves or outcomes observed under current interventions. These designs are particularly useful when researchers want to maximize the number of units receiving effective treatments while still maintaining experimental rigor. However, the current IDEAL survey cannot accommodate studies using an adaptive experimental design.

- **Different analytic samples:** One manuscript analyzes the full experimental sample while another focuses on a subgroup. These represent different analyses of one experiment.
- **Sequential interventions to the same sample:** If a study first implements Intervention A, collects endline data, publishes results in one manuscript, then later implements Intervention B to the same experimental sample and publishes results in a second manuscript, IDEAL codes these manuscripts separately but labels them as **related studies** because they share the same participants. Users should be aware that results from these manuscripts are not independent, a critical consideration for meta-analysis.

3.4 Related Studies from Overlapping Samples

To the extent possible, given the scale of the IDEAL library, IDEAL identifies and flags related studies when different experiments share substantial portions of their samples, even if the interventions or randomization approaches differ. "Substantial overlap" typically means that a majority of participants in one study also participated in the other, though reviewers use judgment based on the specific context.

For instance, if a new experiment is overlaid on an existing RCT's sample, both studies would be coded and flagged as related when this relationship can be determined from the published literature. This transparency about sample overlap, where detectable, enables appropriate statistical methods when synthesizing evidence and helps users avoid treating non-independent samples as independent observations in meta-analyses.

4. IDEAL Coding Components

IDEAL organizes information into five interconnected components that together provide a comprehensive picture of the experimental research: the design structure, the population studied, the interventions tested, the outcomes measured, and the effects estimated. These components were identified through consultation with global standards (including the Data Documentation Initiative and clinical trial registries) and a detailed crosswalk of 15 evidence aggregation instruments used by leading research organizations.

This approach enables diverse users to locate studies relevant to their questions and extract the essential information needed for their purposes. Because IDEAL documents what study authors report without imposing quality ratings or risk-of-bias judgements, users can apply their own frameworks and standards to the data.

While IDEAL does not assign study ratings or make quality determinations, it systematically collects information that enables users to assess risk of bias according to their preferred frameworks. Risk of bias refers to systematic errors in study design, conduct, or analysis that could lead to over- or under-estimation of treatment effects. Key fields that support risk-of-bias assessment include: sample sizes at randomization and analysis, which enable calculation of attrition; randomization method and unit of randomization, which inform assessment of selection bias; and documentation of analytical approaches, which allows evaluation of whether authors appropriately handled design features such as clustering. By capturing this information descriptively, IDEAL allows users to apply [Cochrane-style risk-of-bias](#) judgments, WWC-style threshold criteria, or other frameworks appropriate to their research questions.

4.1 Core Components

1. Experimental Design Structure: IDEAL collects the basic architecture of each experiment, including how many experiments a paper evaluates, what study arms exist within each experiment, and how intervention units were assigned to these arms. Understanding experimental structure is

fundamental to interpreting any findings. It clarifies what comparisons are being made and how treatment assignment was determined. This information allows evidence synthesis across studies with similar designs and enables appropriate statistical methods that account for design features like clustering or stratification.

This component includes:

- Number of distinct experiments in the paper
- Intervention assignment strategy
- Study arm labels and descriptions
- Unit(s) of randomization—the level at which random assignment occurred (individuals, households, classrooms, schools, communities, or other clusters)
- Stratification—whether groups of participants based on baseline characteristics were used in random assignment

2. Study Sample: IDEAL collects characteristics that define who participated in the study, how they were identified and recruited, and at what scale the intervention operated. Population characteristics determine the scope of inference for any study's findings and are essential for assessing external validity. Documenting eligibility criteria, target populations, and sampling approaches enables evidence synthesis across similar populations and supports assessments of whether findings might generalize to other contexts. Recording sample sizes at randomization and analysis provides the foundation for calculating attrition, a key risk-of-bias indicator for RCTs.

This component includes:

- Eligibility criteria (who could participate) and exclusion criteria (who could not)
- Target population description
- Geographic and institutional scale of implementation
- Sampling and recruitment approach
- Sample sizes at randomization and at analysis

3. Interventions: IDEAL collects detailed descriptions of what participants in each study arm experienced, including both treatment interventions and comparison conditions. Comprehensive intervention documentation is necessary for multiple purposes: assessing the feasibility of replication, understanding what aspects of interventions may have driven effects, determining whether interventions across studies are similar enough for evidence synthesis, and evaluating whether findings might transfer to different implementation contexts. Complete descriptions of comparison conditions are equally important, as the contrast between treatment and comparison determines what the effect estimate actually represents.

This component includes:

- Intervention content and components
- Intensity: duration, frequency, dosage, or amount delivered
- Implementation context: geographic location, institutional settings
- Control condition descriptions (a description of what participants in the control group actually experienced)

4. Outcomes and Measurement: IDEAL collects information about outcomes for which treatment effect estimates are reported, including what constructs they measure and how measurement occurred. Detailed outcome documentation enables identification of studies measuring relevant constructs and assessment of construct validity. Documentation of units of analysis and timing clarifies what effects represent.

This component includes:

- Outcome variable definitions
- Measurement instruments and approaches used
- Unit of analysis (which may differ from the unit of randomization)

- Timing of data collection rounds

5. Treatment Effect Estimates: IDEAL collects treatment effect estimates that include both effect magnitude (point estimates) and precision (standard errors, confidence intervals, or p-values), drawn from exhibits in the main text of papers. Treatment effects are the primary output of experiments and the foundation for evidence synthesis. However, studies often report multiple estimates for the same outcome using different samples, analytical approaches, or model specifications. Systematic documentation of which estimates are reported and how they were produced is essential cross-study comparisons and evidence aggregation. IDEAL applies a prioritization approach to ensure comparable estimates across studies while also documenting author preferences, allowing users to examine how analytical choices affect conclusions.

To maintain a manageable coding workload, IDEAL sets limits on which outcomes and follow-up periods are eligible for treatment effect extraction.

- **Follow-up periods.** For outcomes reported across more than two estimation periods (e.g., midline, first follow-up, second follow-up), IDEAL collects a maximum of two periods. Post-treatment completion follow-up periods are always prioritized over periods that fall within the treatment period. When more than two post-treatment completion follow-up periods are available for the same outcome and comparison, IDEAL extracts only the first and the last. When there is exactly one within-treatment period and one post-treatment period, both are collected.
- **Index outcomes.** When an outcome is an index constructed from component outcomes, IDEAL collects both the index and all its components only if two conditions are jointly met: (1) the index and all component outcomes appear in the same table in the main text, and (2) the index contains fewer than 10 components. If only a subset of components is reported alongside the index, or if the index contains 10 or more components, only the index outcome and its associated treatment effects are collected.
- **Studies with a large number of outcomes.** For studies reporting treatment effects for more than 50 outcomes, IDEAL applies the following restrictions in sequence:
 - Outcomes for which the paper does not provide sufficient information to compute standardized effect sizes are excluded.
 - If the paper distinguishes between primary and secondary outcomes, only primary outcomes are retained.
 - If more than 50 outcomes remain after applying the above two restrictions, IDEAL collects the first 50 outcomes in the order they appear in the paper's exhibits.
- **Use of appendix and supplementary materials.** IDEAL extracts information from the paper's appendix or supplementary materials only in two circumstances: (1) the estimate or precision information for IDEAL's preferred specification of an eligible treatment effect is presented exclusively in the appendix, or (2) details required to fully interpret a figure or table in the main text are provided in the appendix.

4.2 IDEAL's approach to selecting treatment effect estimates

To select treatment effect estimates, IDEAL follows a systematic prioritization approach that balances standardization with documentation of study-specific decisions.

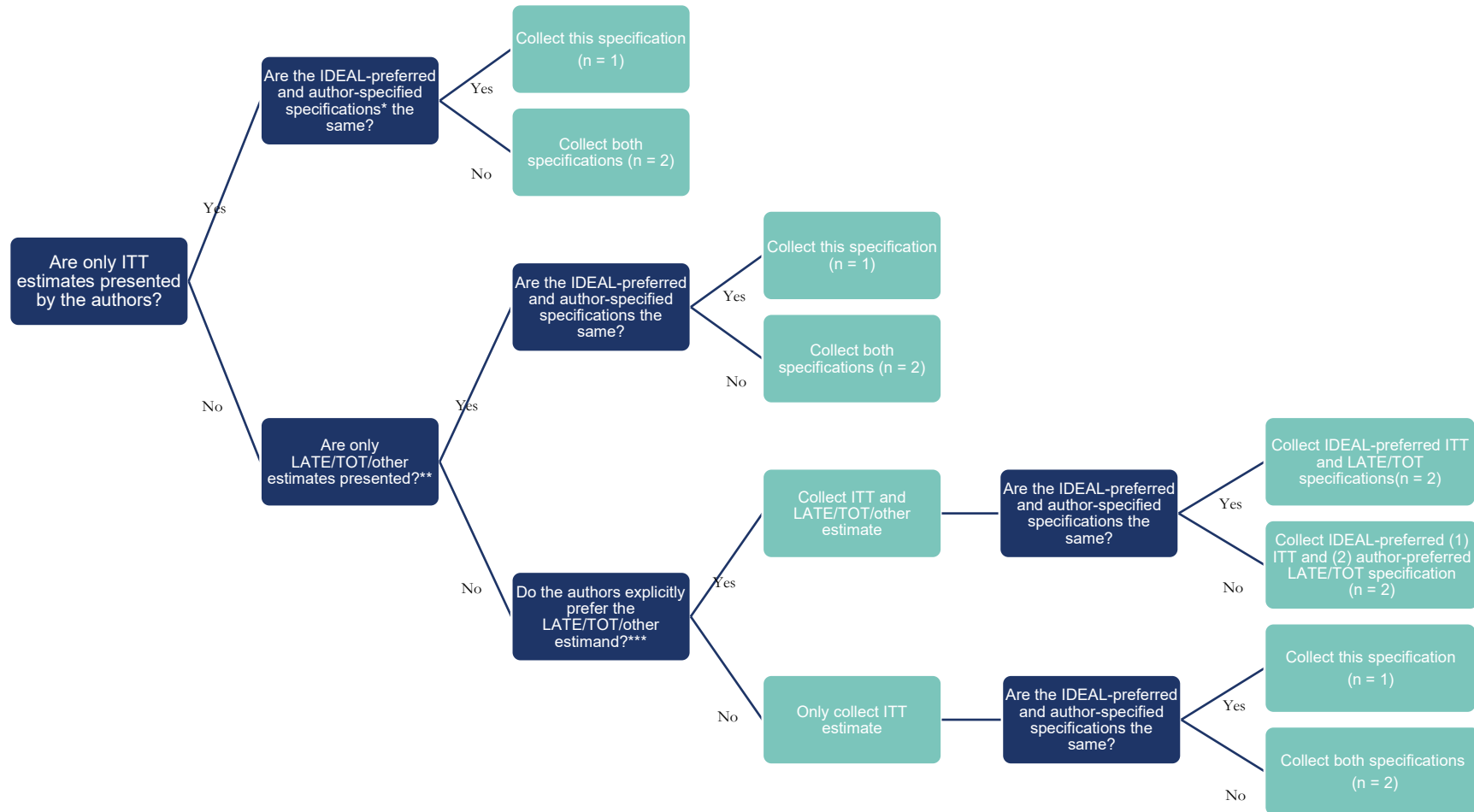
Estimand: IDEAL prioritizes intention-to-treat (ITT) effects. ITT estimates include all participants as originally assigned regardless of whether they received the intervention. These effects answer the policy-relevant question: "What is the impact of offering this intervention?" IDEAL collects Local Average Treatment Effect (LATE), Treatment-on-the-Treated (TOT), or other estimands when ITT estimates are unavailable or when LATE/TOT/other is the authors' preferred estimand. LATE and

TOT measure effects for those who comply with the assignment or those who actually participated, respectively. See Figure 2 for detailed decision guidance.

Sample: IDEAL prioritizes effects for the full experimental sample. Heterogeneous effects (for example, effects separately for boys and girls) are collected when full-sample results are unavailable or when heterogeneous effects are part of the study's primary research questions. See Figure 3 for detailed decision guidance.

Analytical specification: Studies often report multiple specifications—different combinations of statistical controls—for the same outcome. IDEAL selects specifications according to a hierarchy that prioritizes adjusting for design features and baseline outcomes (Table 1): first, specifications with stratification controls (when stratification was used) plus baseline outcome measures; second, specifications with stratification controls only; third, specifications with stratification controls plus other baseline covariates, choosing the simplest one when multiple options exist. See Figure 4 Multiple Specifications Flowchart for detailed decision guidance. When there is more than one estimation model (e.g. OLS and probit) for the same outcome-contrast pair using the same empirical specifications and estimand without a direct preference from the authors, IDEAL will record treatment effect estimated by the linear model (in most times, the OLS regression results). This hierarchy reflects principles from Athey and Imbens (2017), who demonstrate that stratification can only improve—never harm—precision, and that regression adjustment with additional covariates offers uncertain benefits while potentially increasing finite-sample variance when those covariates have limited predictive power. IDEAL therefore favors specifications that reflect design features while minimizing dependence on modeling assumptions.

Figure 2: Decision diagram for IDEAL preferred estimand and empirical specifications



*We distinguish between the IDEAL-preferred estimand (e.g., ITT, LATE) and IDEAL-preferred specifications (e.g., strata fixed effects)

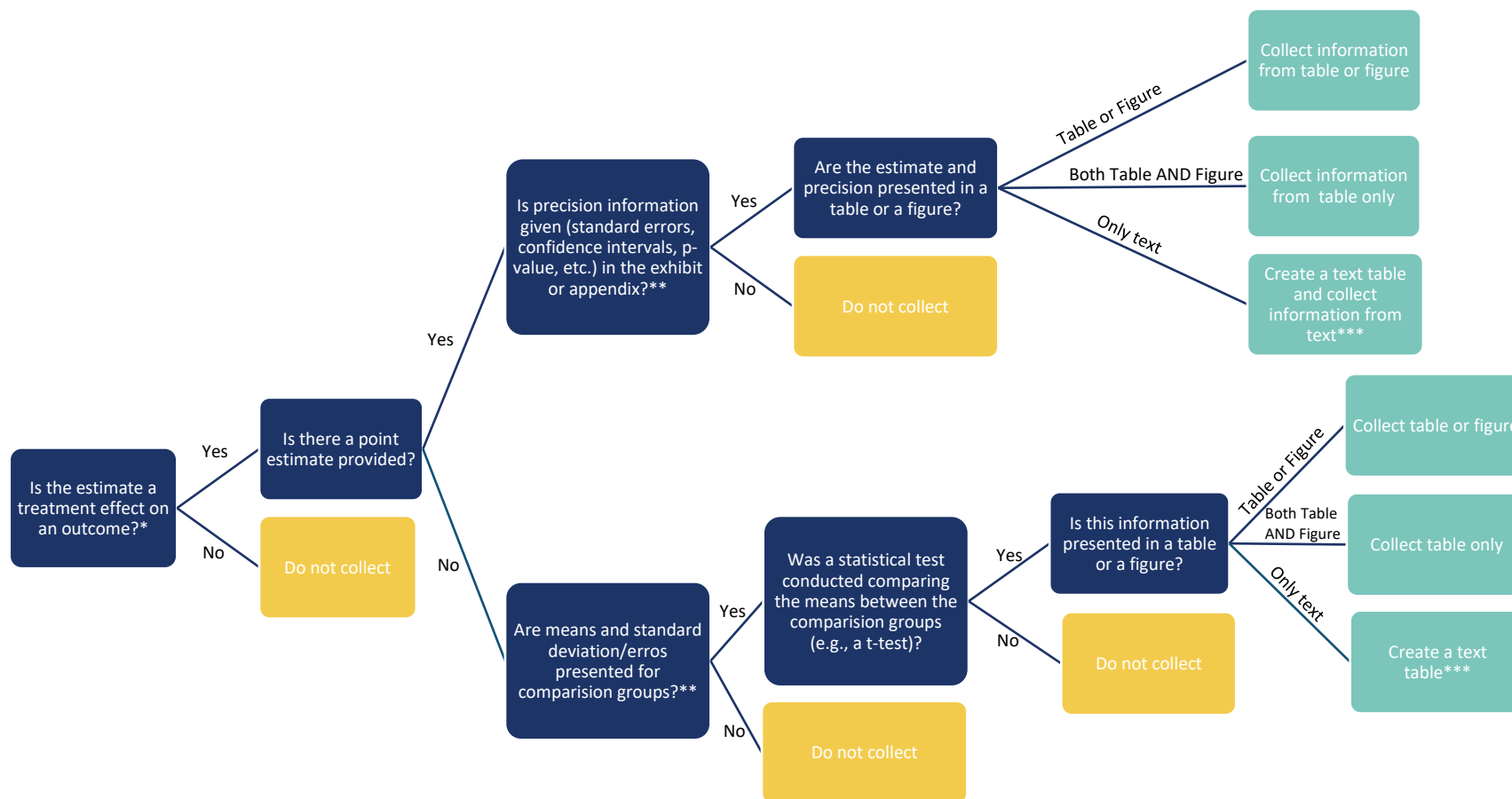
**Even if estimated using 2SLS/IV regression

***Explicit preference: **1)** authors state direct preference for LATE/TOT estimand; **2)** LATE/TOT are only estimate discussed by authors in abstract and introduction; or **3)** LATE/TOT are main research question (e.g., if authors discuss compliance)

If one of the three conditions above is not met, but LATE/TOT estimates are highlighted as topline results (in abstract or introduction), collect LATE/TOT estimates but only for those specific outcomes.

Figure 3: IDEAL eligible treatment effects

When to include an exhibit with treatment effects (for full sample)?



* Note that authors will sometimes include tables or figures illustrating program implementation or take-up. Unless take-up is an explicit outcome noted by the authors, do not include these estimates.

**Note that this information may be included in the text of the main paper or its appendices, especially for figures.

***Some papers may not present estimates in tables or figures but instead discuss them in the text. If this is the case, create a “text table” (e.g., Text table 1) with these estimates.

Table 1: IDEAL Ranking for Empirical Specifications

The following priority rules determine which empirical specifications are collected as IDEAL-preferred:

- Stratification adjustment > No stratification adjustment
- Baseline outcome included > Baseline outcome not included
- No other controls included > Other controls included*
- Other controls: Pre-treatment, static, exogenous controls > Post-treatment, non-static, endogenous controls

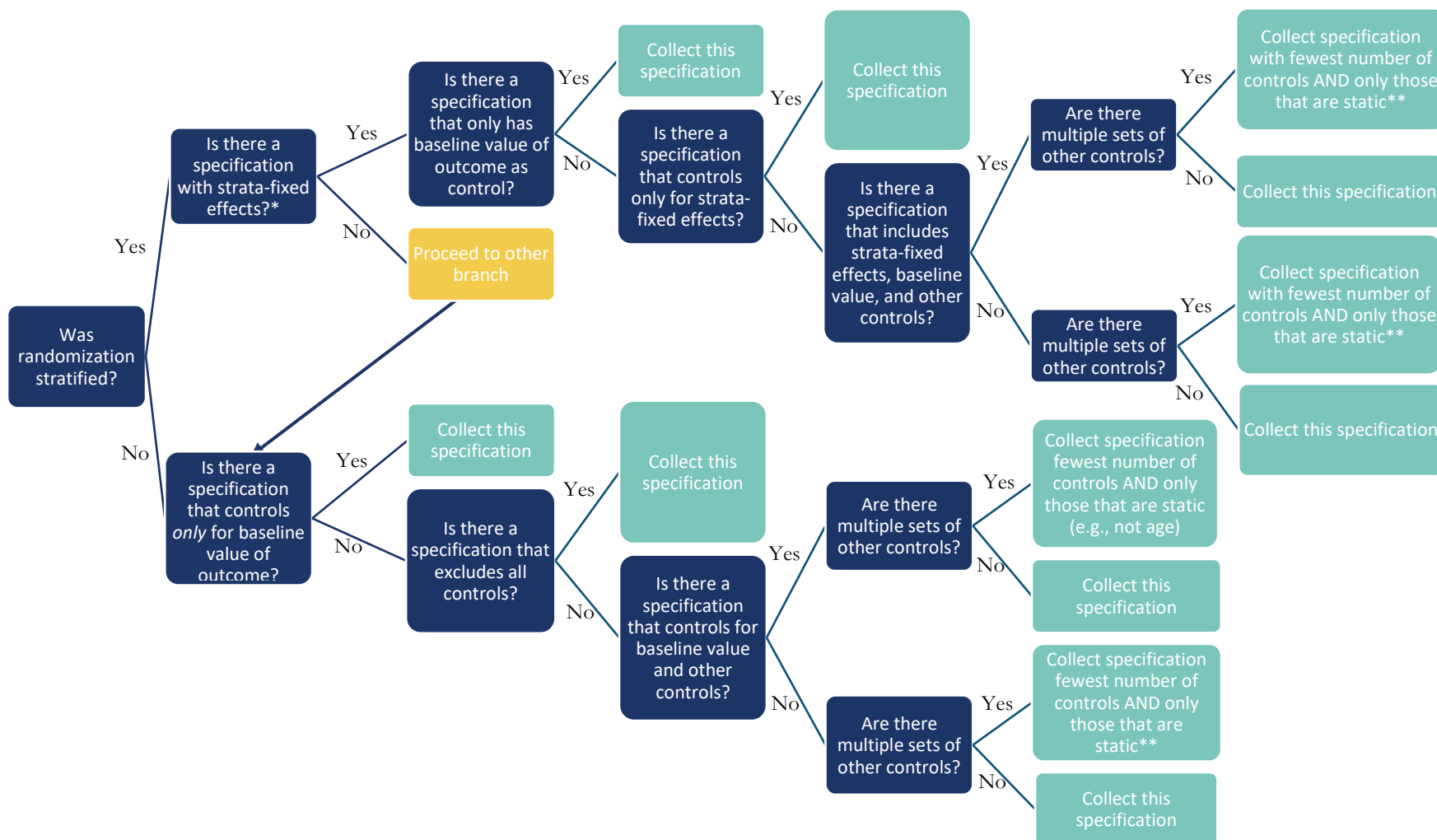
Based on these rules, the IDEAL-preferred specification ranking is as follows, from highest (1) to lowest (8) preferred specification:

Ranking	Strata adjustment	Baseline outcome	Other controls	No other controls
1	x	x		
2	x			x
3	x	x	x	
4	x		x	
5		x		
6				x
7		x	x	
8			x	

*When there are multiple specifications with different sets of other controls, the most parsimonious set is preferred.

Figure 4: IDEAL's preferences over multiple empirical specifications

How to choose IDEAL preferred specifications when there are multiple specifications presented



* Note that some authors will include fixed effects for other units of analysis besides strata. We consider these other controls.

**Controls that are observed before treatment assignment (income at baseline) or that are static over the study period (age, parental education) are preferred over controls that might be affected by treatment assignment (income at endline).

Author preferences: When study authors indicate a preferred estimate or specification that differs from IDEAL's hierarchy, both the IDEAL-preferred and author-preferred estimates are collected. This allows examination of how analytical choices affect conclusions while maintaining standardized estimates across studies.

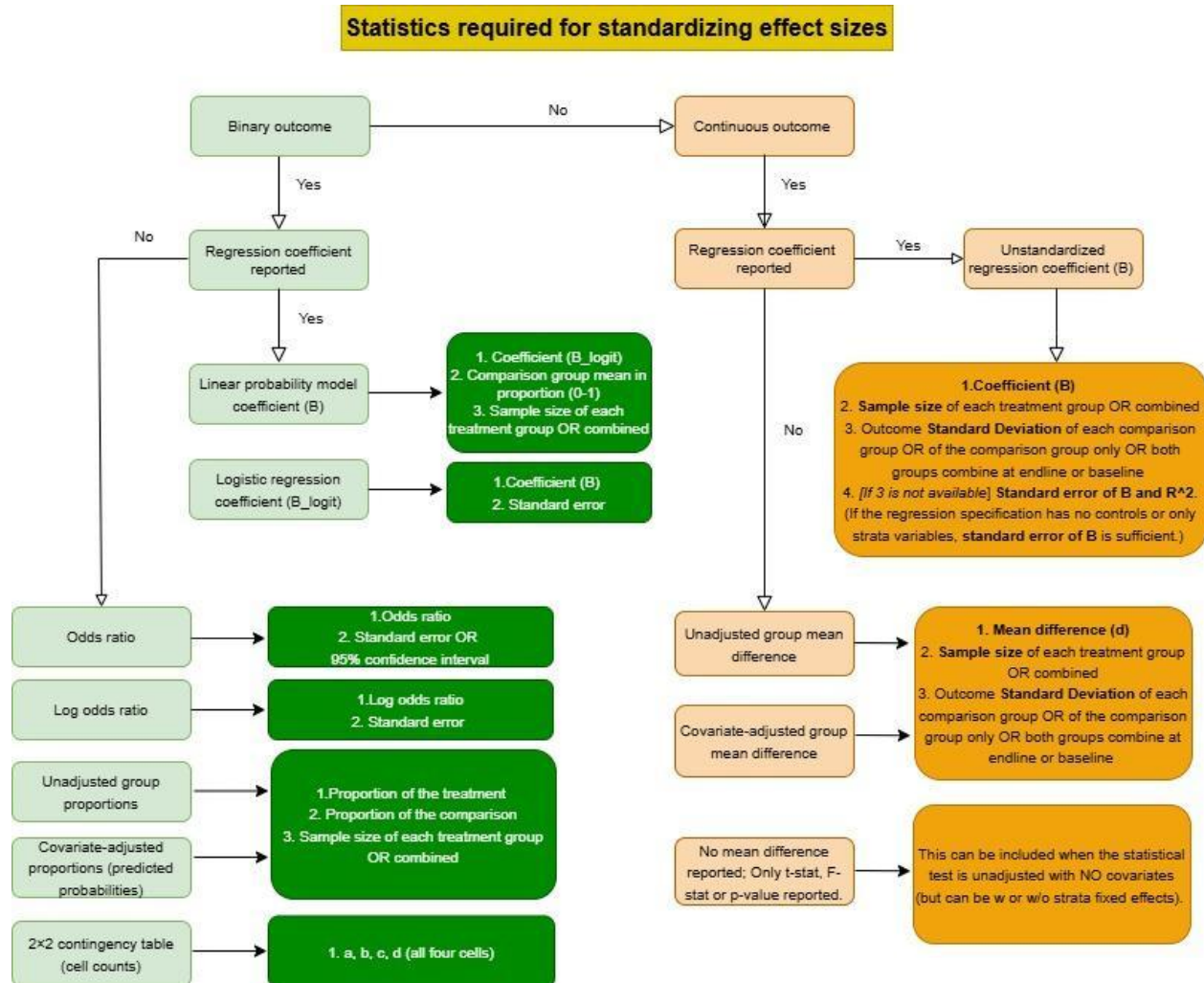
Completeness and location: Eligible treatment effects must appear in the main text (in tables, figures, or narrative) and include both a point estimate and at least one precision measure. Supplementary materials may be consulted to find precision information for effects presented in the main text. IDEAL extracts information from the appendix or supplementary materials only in two cases. The first is when the estimate or precision for IDEAL's preferred specification is not reported in the main text but is found in the appendix. The second is when additional information needed to interpret a figure or table from the main text is provided in the appendix. See Figure 3 for detailed decision guidance.

Heterogeneous Treatment Effects: IDEAL collects HTEs only when they are either (i) the sole treatment effects reported in the study, or (ii) central to the study's primary research questions.

. (i) When HTEs are the sole treatment effects reported in a study, IDEAL treats them as full-sample treatment effects. In these cases, each subgroup outcome should be coded as a separate outcome with a clearly labeled variable name (e.g., *enrollment-female*, *earnings-male*, *yield-low-income households*). IDEAL collects treatment effect estimates for each subgroup, as well as differences between subgroups, provided that the treatment effects for the individual comparison groups can be backed by available effect size and precision estimates. HTEs reported solely through interaction terms, without sufficient information to recover the treatment effects and precision measures for each comparison group, cannot be included in IDEAL.

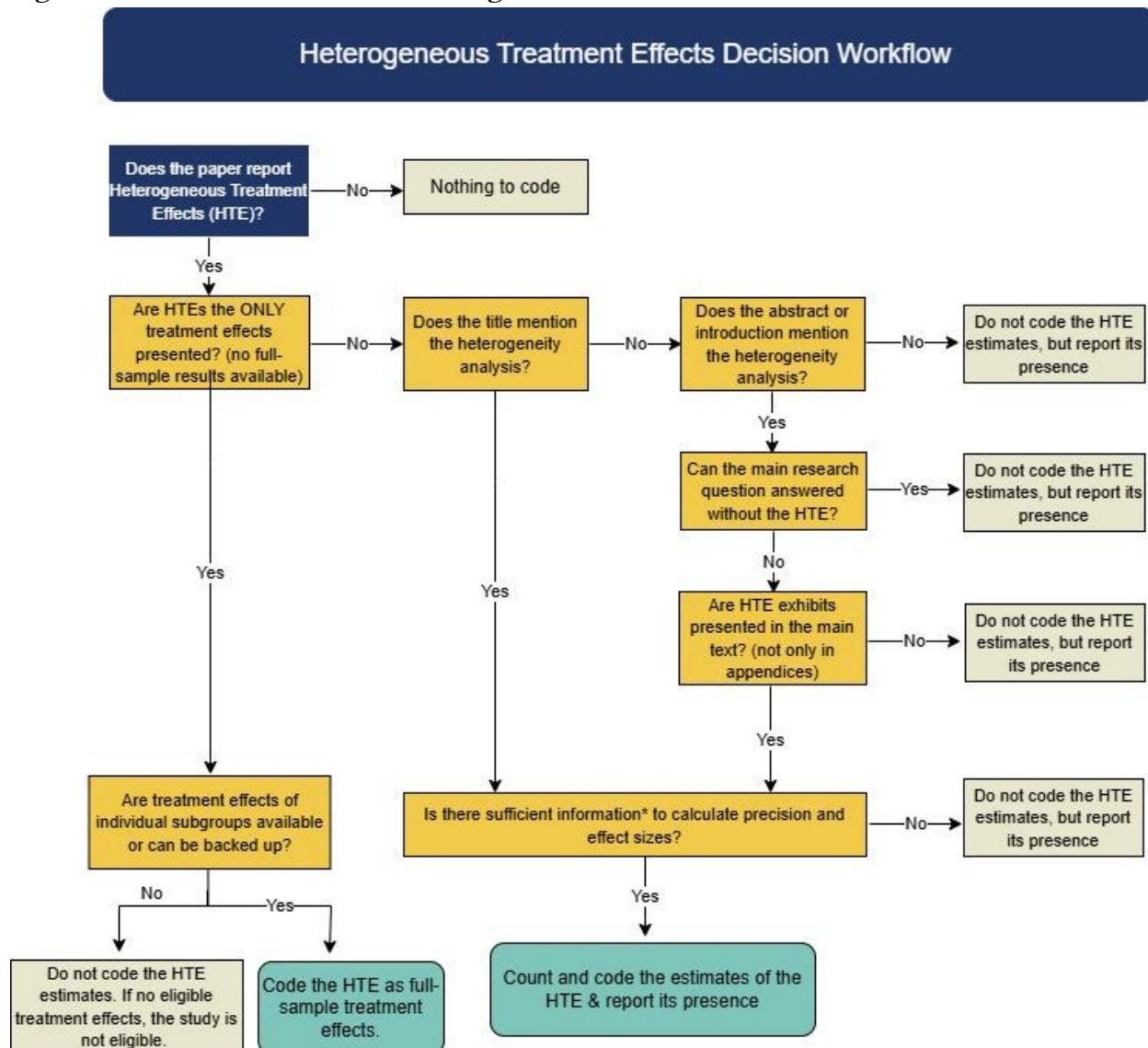
(ii) when both full-sample and HTEs are reported in the study, HTEs are considered primary when mentioned in the title, or when mentioned in the abstract and presented in main text exhibits. For HTEs to be extracted, the paper must report sufficient information for both precision and effect size calculation. For precision: when authors estimate HTEs using separate subsamples, the point estimate and precision (standard error, confidence interval, or p-value) must be reported directly for each subgroup. When authors use treatment-by-subgroup interactions, the reference group effect and its precision are reported directly, but for non-reference groups the point estimate must be calculated by summing the main treatment coefficient and the interaction coefficient — and precision for this combined effect must be reported by the authors, not just for the interaction term alone. For effect size calculation: regardless of estimation approach, the paper must report sufficient information for IDEAL to compute standardized effect sizes for each subgroup (see Figure 5 Effect Size Calculation Workflow for the required inputs under each formula). Spillover effects are treated as main effects, not HTEs, and follow the same collection and standardization requirements as other main effects. See Figure 6 HTE Flowchart for details on when to extract them.

Figure 5: Statistics required for standardizing effects sizes



***If any of the statistics listed in the dark green or dark orange box is not reported, the effect size can not be standardized and should not be included.**

Figure 6: Decision workflow for heterogeneous treatment effects



Notes:

* Sufficient information needs to be reported for the HTE's estimate, precision and standardization.

For estimate and precision:

- If HTEs are estimated using separate subsamples: point estimate and precision (SE, CI, p-value, etc.) reported directly for each subgroup
- If HTEs are estimated using interactions: point estimate and precision reported directly for reference group; for non-reference groups, precision must be reported for the combined effect (main treatment + interaction), not just the interaction term alone

For effect size calculation:

- Subgroup means, subgroup standard deviations, and subgroup sample sizes; OR
- Subgroup treatment effect, its standard error, standard deviation of outcome for that subgroup, and subgroup sample sizes; OR
- For unadjusted analyses only: t-statistic or F-statistic (with numerator df=1) for the subgroup comparison and subgroup sample sizes

Additional note:

This decision workflow is meant for subgroup analysis for example males vs females. It does not apply to spillover effects. Spillover effects will be coded as main effects.

5. The IDEAL Data Extraction Process

IDEAL's data extraction process translates the conceptual components described in Section 4 into a systematic, staged workflow with integrated quality controls. This section provides an overview of how coding is implemented, the rationale for IDEAL's two-stage approach with supervisor checkpoints, and how information flows from paper to database. The process employs strategic double-coding at the final stage where the bulk of detailed information is collected, balancing data quality with coding efficiency. Detailed field-by-field coding protocols and instructions are maintained separately; this section focuses on the overall architecture and logic of the extraction process to help partners and users understand how IDEAL transforms published manuscripts into structured, reusable data.

5.1 Two Stages of Data Extraction

IDEAL uses a two-stage architecture for data extraction rather than a single comprehensive survey. This design serves multiple purposes: it manages the complexity of extracting detailed information from diverse experimental papers, ensures quality through built-in checkpoints before information flows forward, and creates logical dependencies where earlier stages define the scope and structure of later stages. Each stage builds systematically on the previous one, with supervisor review between stages ensuring that only accurate structural information propagates through the workflow. The result is an efficient process that maintains high data quality while avoiding the need for extensive ex-post data cleaning or reconciliation.

Stage 1: Mapping the Experimental Structure and Locating Treatment Effects

Stage 1 first identifies the architecture of what is being tested and where results appear in the manuscript. Coders document the number of experiments in the paper, the interventions and study arms within each experiment, the outcomes for which treatment effects are reported, the empirical specifications used in analyses, the rounds of data collection, and critically, the exhibits (tables, figures, or text sections) that contain treatment effects.

This stage comes first because these structural elements determine what questions appear in later stages. Most importantly, identifying exhibits early in Stage 1 organizes all subsequent coding exhibit-by-exhibit, allowing coders to work systematically through the paper's results table-by-table or figure-by-figure rather than jumping around the manuscript.

The data collection platform includes automated validation checks that prevent logical inconsistencies and data entry errors during all stages. For example, if a coder indicates there are three interventions but only provides labels for two, the system flags this discrepancy. Once the coder completes Stage 1, a supervisor reviews and approves the outputs before the coder can proceed to Stage 2. This early checkpoint is critical because errors in identifying the experimental structure would cascade through all subsequent stages.

The second half of Stage 1 uses the structural information to identify which specific treatment effects exist for each outcome and to determine which estimates are eligible for extraction using IDEAL's preferred specifications. The survey automatically generates questions for all possible combinations of outcomes, arm comparisons, and empirical specifications identified earlier, then presents these questions organized by exhibit.

For example, when a coder reaches Table 3 (identified in Stage 1 as containing treatment effects), the survey asks about each potential treatment effect that could appear in that table: "Does Table 3 contain a treatment effect for math scores comparing Treatment A to Control using Specification 1 for measurement period X?" This exhibit-based organization allows coders to focus on one table or figure at a time, systematically confirming which treatment effects are present.

After completing Stage 1, a supervisor reviews and approves the entered data before the coder proceeds to Stage 2. This checkpoint ensures that only the eligible treatment effects—those actually present in the paper and matching IDEAL's selection criteria—are flagged for detailed extraction in the next stage.

Stage 2: Extracting Detailed Information

Stage 2 collects comprehensive details about the experiment, interventions, outcomes, and treatment effect estimates. This is the most extensive stage, covering the bulk of IDEAL's metadata including intervention descriptions (content, intensity, delivery mechanisms, implementation context), outcome definitions (measurement instruments, units of analysis, type), sample characteristics (eligibility criteria, sample sizes at randomization and analysis), and treatment effect values (point estimates, standard errors, confidence intervals, p-values, sample sizes used in estimation).

When extracting treatment effect estimates, questions are organized by exhibit, mirroring the structure from Stage 1. For each treatment effect identified in Stage 1, coders enter the numerical values for the magnitude of the effect, precision measures and sample sizes required for standardization. All collected data is automatically associated with the correct experiments, arms, outcomes, and specifications identified in previous stages, eliminating the need for ex-post linking or matching.

Because Stage 2 captures the most detailed information and involves numerical data entry where errors could occur, this stage undergoes double-coding. Two independent coders complete Stage 2 for each paper, with discrepancies between the two coded entries resolved through supervisor review (described in Section 6).

Data Flow and Integration

Information flows seamlessly across the two stages through the data collection platform preloading functionality. Data collected in Stage 1—such as intervention labels, outcome names, and exhibit identifiers—automatically populates the relevant questions in Stage 2. Similarly, the treatment effects confirmed in Stage 1 determine which detailed extraction questions appear in Stage 2. By the end of Stage 2, all data resides in one integrated relational dataset with clear linkages between structural elements, treatment effect locations, and detailed estimates.

5.2 The Survey Platform

IDEAL is using SurveyCTO and a custom survey website as its data collection platforms in the pilot stage, chosen for its accommodation of ODK tools that are widely used in development research and fieldwork. The platform's automated validation catches common errors in real-time during Stages 1 and 2—for example, flagging if a coder indicates three interventions but only labels two, or if responses create logical inconsistencies. Dynamic survey logic ensures that questions in later stages automatically populate based on earlier responses, so Stage 2 generates questions for each outcome identified in Stage 1, and Stage 3 presents extraction prompts for each treatment effect confirmed in Stage 2.

The platform also supports supervisor review and approval workflows between stages, preventing coders from proceeding until their work has been validated. Detailed survey forms translate IDEAL's conceptual framework into structured questions with appropriate skip patterns, validation rules, and consistency checks, all designed to facilitate coding, minimize errors and ensure data quality.

6. Quality Assurance

IDEAL employs a multi-layered quality assurance system to ensure accurate and consistent data extraction across all coded papers. The approach combines comprehensive training, built-in

validation checks during coding, structured supervisor review between coding stages, double-coding of detailed information in Stage 2, and systematic calculation of quality metrics. This section describes each component of IDEAL's quality assurance framework and the metrics used to monitor and improve coding quality during the pilot phase.

6.1 Training and Preparation

Coders are hired through a competitive process with a screening coding exercise. Before coding live papers, all coders complete a comprehensive training program. The training package includes detailed coding protocols, training slides and videos for concepts and fields prepared by Principal Investigators, and training videos demonstrating how to complete surveys using the survey mask on the designated platform.

During training, coders complete practice papers to develop proficiency with the coding system. For these training papers, supervisors extract the ground truth—the correct answers for each field—which is then reviewed by PIs to ensure accuracy. This ground truth serves two purposes: providing guidance to coders as they learn the system and calculating accuracy metrics to assess coder readiness.

Throughout the training period and during coding periods, weekly office hours sessions are hosted to answer coders' questions and provide additional support.

6.2 Built-in Quality Controls During Coding

The SurveyCTO platform incorporates automated validation checks and skip logics in all stages of data extraction to prevent errors in real-time. These checks catch logical inconsistencies, such as when a coder indicates there are three treatment arms but only provides labels for two, or when responses create contradictions across related fields. They also streamline coding by skipping irrelevant questions based on the previous answers to related fields. For example, the mapping of interventions to study arms will be skipped if the assignment strategy is parallel as a study arm will be automatically generated for each intervention. These built-in controls reduce coding errors at the point of data entry and improve efficiency by alerting coders to problems immediately rather than discovering them during later review stages.

6.3 Supervisor Review Process

IDEAL implements two rounds of quality checks during the coding process for each paper: after Stage 1 and after Stage 2. These supervisor reviews serve dual purposes—building a dataset with correct answers for the next coding stage and providing feedback to coders for continuous improvement.

In Stage 1, the supervisor reviews the information in each field. The supervisor can correct minor errors or provide feedback to the coder and request resubmission of Stage 1 if major errors are made in the coding. When revisions are submitted, the review process starts over.

Supervisor reviews are built into the coding workflow through SurveyCTO. Coders can only progress to the next stage after the supervisor completes and approves the review, ensuring that only validated information flows forward.

While Stage 1 relies only on supervisor review for quality control, Stage 2 employs double-coding. Two independent coders complete Stage 2 for each paper, extracting detailed information without seeing each other's work. Because Stage 2 captures the bulk of IDEAL's metadata—intervention descriptions, outcome definitions, sample characteristics, and treatment effect values—double-coding at this stage ensures reliability of the final data while maintaining efficiency in earlier stages.

After both coders complete Stage 2, supervisors compare the two sets of coding, identify discrepancies, and select or enter the preferred values that enter the dataset.

6.4 PI Spot Check

An additional quality assurance layer occurs through PI spot checks. PIs from each organization conduct spot checks on a random subset of fields and papers after supervisor review. Initially, PIs check 20% of fields for 20% of randomly selected papers; this may be reduced to 10% of fields later in the pilot as quality stabilizes.

If a PI flags an error or has comments during the spot check, the supervisor responsible for that paper corrects the corresponding data in the surveys. The PI spot check is conducted through a standalone survey on SurveyCTO and serves as an additional quality assurance measure independent of the main coding workflow.

6.5 Quality Metrics

IDEAL calculates two primary quality metrics during the pilot phase: accuracy and reliability. These metrics use data from supervisor checks and PI spot checks and are calculated at multiple levels to provide detailed insights into coding quality.

The accuracy rates will be calculated using the comparison between the supervisor checked data and the PI cleared data. Therefore, the statistics will be based on the set of papers that go through a PI check.

Accuracy. A given entry is considered accurate if it represents what is described in the manuscript and is verified by a PI for information that requires inference. Accuracy is calculated at several levels:

Accuracy of a paper = (Number of fields verified accurate) / (Total number of fields in the paper)

Accuracy of a section per paper = (Number of fields verified accurate in the section) / (Total number of fields in the section)

Average accuracy of a coder = Sum of accuracy of a paper across all papers coded / (Total number of papers coded by that coder)

Reliability. A given entry is considered reliable if two coders extract the same information from the paper in the same way. Importantly, two coders could extract information in different ways and still both be accurate. For example, defining interventions as "training" and "cash" versus "training" and "training plus cash" could both accurately describe a trial but would not be coded identically.

IDEAL aims for coders to be both accurate and consistent with each other.

Given the IDEAL multi-stage coding workflow, reliability for Stage 1 and Stage 2 fields is calculated by comparing the entries between a coder and a supervisor. For Stage 3, when double coding is performed by two coders, the reliability is calculated based on the entries by two coders.

Reliability is calculated as:

Reliability of a field = (Number of double-coded extractions scored as reliable) / (Total number of unique papers coded)

[PLACEHOLDER: Information needed on target reliability thresholds, if any exist, and how reliability data is used to identify training needs or problematic fields.]

[PLACEHOLDER: Appendix table to list the fields to be included in the calculation of accuracy and reliability.]

6.6 Ongoing Quality Monitoring

- PI-cleared data processing to check and remove outliers.
- Feedback request form available on the IDEAL data platform.

[PLACEHOLDER: Information needed on processes for ongoing quality monitoring beyond the formal metrics—spot checks of completed papers, tracking patterns in discrepancies, identifying fields needing clearer guidance, iterative improvements to training and protocols based on pilot experience.]

Reference:

Athey, S., & Imbens, G. W. (2017). The econometrics of randomized experiments. In Handbook of Economic Field Experiments (Vol. 1, pp. 73-140).